$$\int_{-1}^{1} f(x)\,dx \approx \sum_{i=1}^{n} w_i f(x_n)$$

(n=4):

$w_1 = w_4$, $w_2 = w_3$

$x_3 = x_2$, $x_1 = -1$, $x_4 = 1$

(x>0)

$f(x)=1$:

$$\int_{-1}^{1} 1\,dx = 2$$

$$2w_1 + 2w_2 = 2 \Rightarrow w_1 + w_2 = 1$$

$f(x)=x^2$:

$$\int_{-1}^{1} x^2\,dx = \frac{2}{3} \Rightarrow$$

$$\Rightarrow 2w_1 + 2w_2 x^2 = \frac{2}{3} \Rightarrow w_1 + w_2 x^2 = \frac{1}{3} \Rightarrow w_2(x^2 - x^1) = \frac{2}{15}$$

$$\Rightarrow w_2(1 - x^2) = \frac{2}{5}$$

$$\Downarrow$$

$$x^2 = \frac{1}{5} \Rightarrow x = \frac{1}{\sqrt{5}}$$

$f(x)=x^4$:

$$\int_{-1}^{1} x^4\,dx = \frac{2}{5}$$

$$2w_1 + 2w_2 x^4 = \frac{2}{5} \Rightarrow w_1 + w_2 x^4 = \frac{1}{5}$$

$$w_2\left(1 - \frac{1}{5}\right) = \frac{2}{3} \Rightarrow w_2 \cdot \frac{4}{5} = \frac{2}{3} \Rightarrow w_2 = \frac{5}{6}$$

$$w_1 = 1 - \frac{5}{6} = \frac{1}{6}$$

| Узел | $-1$ | $-\frac{1}{\sqrt{5}}$ | $\frac{1}{\sqrt{5}}$ | $1$ |
|---|---|---|---|---|
| Вес | $\frac{1}{6}$ | $\frac{5}{6}$ | $\frac{5}{6}$ | $\frac{1}{6}$ |